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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER - Discipline-specific Core Course (DSC)-MG4DSCMBG202

APPLIED MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A - Short Notes Questions - Answer any 10 questions- ***Each question carries 2 marks***

1. List out fermented milk products. **[K] / [CO1]**

- Cheese, Kefir, Buttermilk, Yoghurt, etc. Write any 3

2. Name the enzyme involved in nitrogen fixation [K] / [CO2]

- Nitrogenase enzyme

3. Name 2 commonly occurring bacterial genera in marine water [K] / [CO3]

- Alteromonas, Vibrio sps, Flavobacterium, etc.

4. Write a note on miso **[K] / [CO1]**

- Miso is an Oriental fermented food. fermenting soybeans - Culture used: Aspergillus oryzae.

5. Name two fungi that can be used as SCP. **[K] / [CO1]**

- *Saccharomyces cerevisiae, Candida utilis, Fusarium venenatum, and Aspergillus niger, etc.*

6. Name the biofertilizer which helps in fixing atmospheric nitrogen in leguminous plants.

- Rhizobium [K]/[CO2]

7. What is bioinsecticide [K] / [CO2]

- A Bioinsecticide is any microbial, plant-based, or natural substance used to control insect pests in agriculture. Eg: *Bacillus thuringiensis, Beauveria bassiana.*

8. What is the littoral zone? **[K] / [CO3]**

- The Littoral zone is the shallow region near the shore where sunlight can penetrate to the bottom, allowing growth of rooted plants and algae.

9. Name two useful microbes in food production. **[K] / [CO1]**

- Saccharomyces cerevisiae, Lactobacillus, Aspergillus oryzae, Penicillium, etc.

10. Name 2 protozoan diseases that can be transmitted through contaminated water. [K] / [CO3]

- Amoebiasis caused by *Entamoeba histolytica,* Balantidiasis caused by *Balantidium coli,* Primary Amoebic Meningoencephalitis caused by *Naegleria fowleri, etc.*

11. Give examples of foodborne illness. [K] / [CO1]

- Salmonellosis, Botulism, Listeriosis, etc.

12. Give 2 examples of airborne fungal infections. [K] / [CO4]

- Aspergillosis, cryptococcosis, Histoplasmosis, etc.

Part B- Short Essay Questions- Answer any 4 questions - *Each question carries 5 marks*

13. Explain mutualism in soil microorganisms. [U] / [CO2]

- Mutualism is a type of symbiotic relationship in which both organisms benefit from the association. Explain any two mutualistic relationships – Nitrogen fixation, mycorrhizal association, etc.

14. Explain the role of microbes in beer production. [U] / [CO1]

- Beer is produced by fermentation of cereal grains (mainly barley) sugars by microorganisms. Explain – Malting, mashing, boiling with worts with hops. Explain the microbes involved in Fermentation, Aging and maturation.

15. Define planktons. Explain phytoplanktons and zooplanktons with examples. [U] / [CO3]

- Plankton refers to the free-floating microorganisms. They must live in open water for at least part of their life. Phytoplankton produce their own food using sunlight. Eg: Diatoms, Chlorella, Anabaena. Zooplankton are animal-like planktons that depend on other organisms for food - protozoans, which feed on phytoplankton, bacteria and detritus. Eg: Amoeba, Paramecium.

16. Describe the advantages and limitations of microbial biopesticides. [U] / [CO2]

- Microbial biopesticides are microorganisms such as bacteria, fungi, viruses, or protozoa used to control pests. Advantages: Environmentally Safe, Target Specificity, Safe to Humans and Animals, Reduced Resistance Development, Eco-friendly Pest Management. Disadvantage: Slow Action, Narrow Spectrum, Environmental Sensitivity, Short Shelf Life.

17. Explain different impaction methods of air sampling. [U] / [CO4]

- Impaction methods are techniques used to collect airborne microorganisms by forcing air onto a solid or semi-solid surface, where particles impact and get trapped.

Explain any two methods - Hollander and Dalla Valle Sampler, Slit sampler, Sieve Sampler.

18. Describe the most probable number(MPN) method for detecting coliforms in milk. [U] / [CO1]

- The MPN method is a technique used to estimate the number of viable coliform bacteria in a milk sample based on growth in liquid media. Coliforms ferment lactose and produce acid

and gas, which is used as an indicator. Explain the method briefly – Presumptive test, confirmatory test and completed test.

Part C -Essay Question- Answer any 1 questions-*Each question carries 10 marks*

19. Explain the different types of spoilage of milk. [K] / [CO1]

- Milk is highly nutritious and easily spoiled by microorganisms. Spoilage results in undesirable changes in taste, odour, texture, and appearance.[2] Explain different spoilage in detail: [8] Acidic Spoilage (Souring), Gas Production (Stormy Fermentation), Proteolytic Spoilage (Putrefaction) , Ropiness / Sliminess, Lipolytic Spoilage (Rancidity), Sweet Curdling - Caused by spore-formers like Bacillus. Milk curdles without much acid production., Flavour changes, Discolorations etc.

20. Explain the zonation of lakes with a neat labelled diagram and describe the microbial communities in lakes. [U] / [CO3]

- Lakes and ponds constitute freshwater habitats - lentic habitats. Limnology. [2] Thermal Stratification - epilimnion, metalimnion or thermocline, and hypolimnion. [2]. Zonation – Explain each zone with a neat diagram - light compensation level, limnetic zone, littoral zone, Pelagic zone, euphotic zone, profundal zone, benthic zone. [4]. Microbial Communities [2]